

Architecture Selection – Urban Rooftop Gardening Co-op App

Architecture Selection

We chose Layered Architecture for the Urban Rooftop Gardening Co-op App.

Architectural Choice

The application can be organized into separate layers for presentation, business logic, and data management. This provides a clear separation between the user interface, gardening and trading operations, and stored community data.

Reason 1 – Clear Separation of Functionality

The app contains different functions such as rooftop plot management, crop management, harvest management, soil knowledge sharing, and surplus produce trading. A layered structure separates these responsibilities, making the system easier to understand and maintain.

Reason 2 – Centralized Data Management

The application needs to manage information about rooftop plots, crops, harvests, soil-maintenance tips, and surplus produce. A data layer provides a centralized way to store and retrieve this information while keeping it separate from the application's business logic.

Security Advantage

Security controls can be implemented in the application and data layers to restrict unauthorized access to community, gardening, plot, and trading information. This helps protect the gardening and community data required by the application.

Performance Benefit

The application can optimize the data and business layers separately for frequently accessed information such as surplus-produce listings. This can help reduce unnecessary processing and support timely updates of available produce listings.